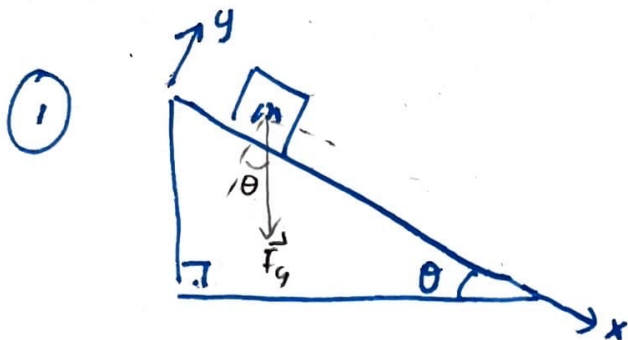


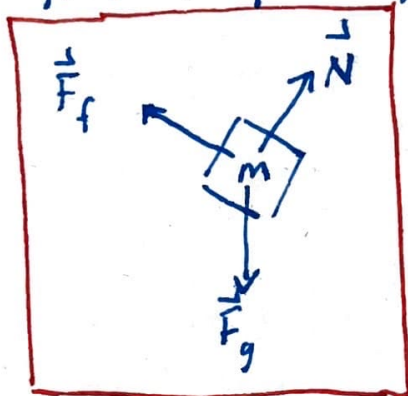
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02MECH - Final



A block of mass m is sliding down on an inclined plane at an inclination angle θ and kinetic friction coefficient μ_k .

a) Draw the free body diagram of the block



$\vec{F}_f$: frictional force
 $\vec{F}_g$: gravitational force
 $\vec{N}$: normal force

b) Write down Newton's second law for y -direction in magnitude form in terms of N , m , g (gravitational constant) and θ .

$$N = mg \cos \theta$$

~~c) Write down Newton's second law for x -direction in magnitude form~~

~~$$mg \sin \theta =$$~~

c) Write down the magnitude of the frictional force acting on the block while it is sliding down in terms of μ_k , m , g , and θ .

$$F_f = N \mu_k = \mu_k mg \cos \theta$$

d) Write down Newton's 2nd law for x -direction in terms of $\ddot{x}$, g , θ and μ_k .

$$mg \sin \theta - \mu_k mg \cos \theta = m \ddot{x} \rightarrow \ddot{x} = g (\sin \theta - \mu_k \cos \theta)$$

e) Solve the differential equation (Newton's 2nd law) you wrote down in (d). and obtain $x(t)$ for the initial conditions $x(0) = 0$, $\dot{x}(0) = 0 = v(0)$

$$\ddot{x} = g(\sin\theta - \mu_k \cos\theta)$$

$$\int d(\dot{x}) = \int g(\sin\theta - \mu_k \cos\theta) dt + C$$

$$\dot{x}(t) = v(t) = g t (\sin\theta - \mu_k \cos\theta) + C$$

Inserting the initial cond: $v(0) = C = 0 \Rightarrow C = 0$

$$\int dx = \int g t (\sin\theta - \mu_k \cos\theta) dt + D$$

$$x(t) = \frac{1}{2} g t^2 (\sin\theta - \mu_k \cos\theta) + D$$

Inserting the initial cond: $x(0) = D = 0 \Rightarrow D = 0$

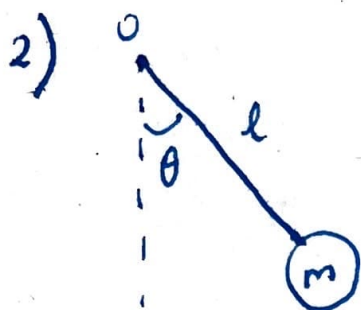
$$x(t) = \frac{1}{2} g t^2 (\sin\theta - \mu_k \cos\theta)$$

Calculate

f) ~~Calculate~~ the position of the block at $t = 2s$ ~~for~~ ~~the~~ for $\theta = 30^\circ$. ($g \approx 10 \text{ m/s}^2$, $\sin 30^\circ = 1/2$, $\mu_k = 1/2\sqrt{3}$, $\cos 30^\circ = \sqrt{3}/2$)

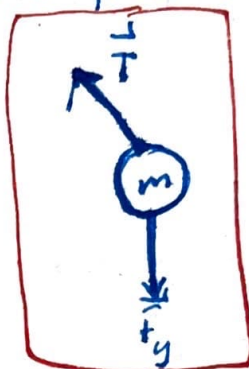
$$x(2) = \frac{1}{2} \cdot 10 \cdot 2^2 \left(\frac{1}{2} - \frac{1}{2\sqrt{3}} \cdot \frac{\sqrt{3}}{2} \right) = 10 \cdot 2 \cdot \frac{1}{4} = 5 \text{ m}$$

$$x(2) = 5 \text{ m}$$



A simple pendulum of mass m and length l is swinging about its equilibrium point $\theta = 0$ with ~~$\theta \ll 1$ (small angle approximation)~~.

a) Draw the free body diagram of mass m .



$\vec{T}$: tension

$\vec{F}_g$: gravitational force

b) What is the rotational inertia I of mass m with respect to the hanging point O in terms of m and l .

$$I = ml^2$$

c) What is the torque N acting on m with respect to O in terms of l , m , g , and θ ?

$$N = lmg \sin \theta$$

$$\vec{N} = \vec{r} \times \vec{F}$$

d) First write down Newton's 2nd law for rotations in terms of N , I and α (angular acceleration

$\ddot{\theta} \equiv \dot{\omega} \equiv \alpha$). And then plug in the results you found in (b) and (c) into Newton's 2nd law. (Note: be careful about the signs)

$$N = I\alpha$$

$$lmg \sin \theta = -ml\ddot{\theta}$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

(Newton's 2nd law)

e) Solve the differential equation you found in (d) to obtain $\theta(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$ for small angle approximation $\theta \ll 1$.

(Given: $\sin \theta \approx \theta$ when $\theta \ll 1$, solution assump. $\theta(t) = e^{\lambda t}$ and $\omega_0^2 = g/l$: angular frequency)

for $\theta(t) = e^{\lambda t}$, $\dot{\theta} = \lambda e^{\lambda t}$, $\ddot{\theta} = \lambda^2 e^{\lambda t}$

$\ddot{\theta} + \frac{g}{l} \theta = 0$

$\lambda^2 e^{\lambda t} + \omega_0^2 e^{\lambda t} = 0 \rightarrow \lambda^2 = -\omega_0^2 \Rightarrow \lambda_{1,2} = \pm i\omega_0$

$\theta_1(t) = e^{i\omega_0 t}$

$\theta_2(t) = e^{-i\omega_0 t}$

$\theta(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$

f) Given that $\theta(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}$ can be written as $\theta(t) = C \cos(\omega_0 t)$, determine C for the initial condition $\theta(0) = \theta_0$.

$\theta(0) = C = \theta_0 \rightarrow \theta(t) = \theta_0 \cos(\omega_0 t)$

g) By using $\theta(t)$ with the initial condition $\theta(0) = \theta_0$, calculate the angular velocity of the pendulum $\omega = \dot{\theta}(t)$.
 ~~Then calculate the kinetic energy of the pendulum when it is at its maximum.~~
 At what time $\dot{\theta}(t)$ take its largest value?
 Calculate the max k.e. of the pendulum. (for the first time)

$\dot{\theta}(t) = -\theta_0 \omega_0 \sin(\omega_0 t)$

$\frac{d\dot{\theta}}{dt} = -\theta_0 \omega_0^2 \cos(\omega_0 t) = 0 \Rightarrow t$ must be $t = \frac{\pi}{2\omega_0}$

$\dot{\theta}(\frac{\pi}{2\omega_0}) = -\theta_0 \omega_0 \sin(\omega_0 \frac{\pi}{2\omega_0}) = -\theta_0 \omega_0$

$T_{\max} = \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{ds}{dt} \right)^2 = \frac{1}{2} m \left(l \frac{d\theta}{dt} \right)^2 = \frac{1}{2} m l^2 \dot{\theta}^2 = \frac{1}{2} m l^2 \theta_0^2 \omega_0^2$

$T_{\max} = \frac{1}{2} m l^2 \theta_0^2 \omega_0^2$

3) Find the force law $f(r)$ for a central force field that allows a particle to move in a spiral orbit given by $r = k\theta^2$, where k is a constant.

$$\text{Given: } \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{m}{\ell^2} r^2 f(r)$$

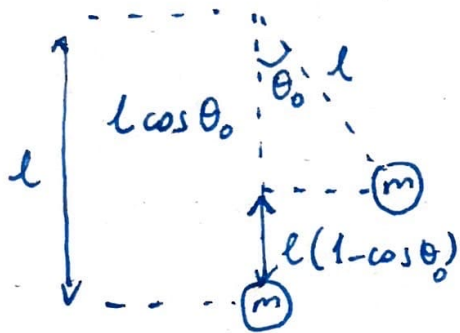
$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) = \frac{d}{d\theta} \frac{d}{d\theta} \left(\frac{1}{k\theta^2} \right) = \frac{1}{k} (-2) \frac{d}{d\theta} \left(\frac{1}{\theta^3} \right) = \frac{6}{k\theta^4} = \frac{6k}{k\theta^2 k\theta^2} = \frac{6k}{r^2}$$

$$\frac{6k}{r^2} + \frac{1}{r} = -\frac{m}{\ell^2} r^2 f(r)$$

$$\Rightarrow f(r) = -\frac{\ell^2}{m} \left(\frac{6k}{r^4} + \frac{1}{r^3} \right)$$

[compare your result with that of (g)]

- h) Calculate the max K.E. of the pendulum from ~~energy~~ conservation of the energy point of view. Assume that mass m has no potential energy ~~when~~ at ~~pendulum~~ the equilibrium point ($\theta=0$).
~~calculate~~ First calculate the potential energy when $\theta = \theta_0$. This ^{potential} energy will be converted to K.E when $\theta = 0$. (Given: $\cos \theta_0 \approx 1 - \frac{\theta_0^2}{2}$ for $\theta_0 \ll 1$)



$$mgl(1 - \cos \theta_0) = \frac{1}{2}mv^2$$

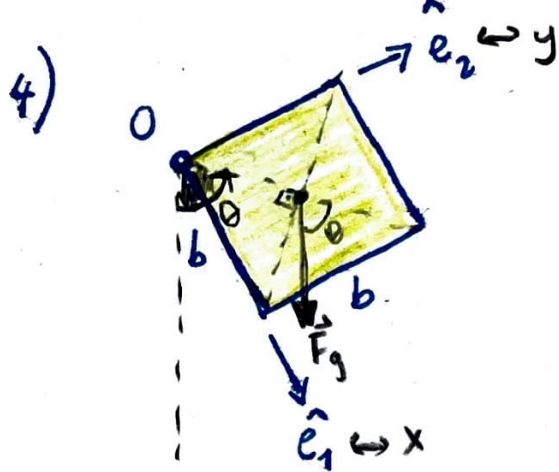
$$2gl(1 - \cos \theta_0) = v^2$$

$$2gl\left(1 - 1 + \frac{\theta_0^2}{2}\right) = v^2$$

$$v^2 = gl\theta_0^2$$

$$T_{\max} = \frac{1}{2}mv^2 = \frac{1}{2}mgl\theta_0^2 = \frac{1}{2}m\left(\frac{g}{l}\right)l^2\theta_0^2 = \frac{1}{2}ml\theta_0^2\omega_0^2$$

$$T_{\max} = \frac{1}{2}ml^2\theta_0^2\omega_0^2$$



with side length b
 An homogeneous square plate is hung from one of its corners O . The area ~~surface~~ density of the plate is $\rho_A = \frac{M}{b^2}$ where M is the total mass of the plate. We assume that the plate is effectively 2 dimensional. There is no mass dist in z -direction.

a) Calculate the inertia tensor ~~by using~~ $[I]$ by using

$$I_{ij} = \int \rho(r) \left(\delta_{ij} \sum_k x_k^2 - x_i x_j \right) dV$$

$z=0$ (no mass dist. in z dir.)

$$I_{11} = \frac{M}{b^2} \int_0^b \int_0^b (y^2 + z^2) dx dy = \frac{M}{b^2} b \frac{b^3}{3} = \frac{1}{3} M b^2$$

$$I_{22} = \frac{M}{b^2} \int_0^b \int_0^b (x^2 + z^2) dx dy = \frac{1}{3} M b^2$$

$$I_{33} = \frac{M}{b^2} \int_0^b \int_0^b (x^2 + y^2) dx dy = \frac{M}{b^2} \left\{ \iint x^2 dx dy + \iint y^2 dx dy \right\}$$

$$= \frac{M}{b^2} \left\{ b \frac{b^3}{3} + b \frac{b^3}{3} \right\} = \frac{M}{b^2} 2 \frac{b^4}{3} = \frac{2}{3} M b^2$$

$$I_{12} = -\frac{M}{b^2} \int_0^b \int_0^b xy dx dy = -\frac{M}{b^2} \frac{b^2}{2} \frac{b^2}{2} = -\frac{1}{4} M b^2$$

$$I_{13} = 0 \quad (\text{since } z=0)$$

$$I_{23} = 0 \quad (\text{since } z=0)$$

$$\text{Let } M b^2 \equiv \epsilon$$

$$[I] = \begin{bmatrix} \epsilon/3 & -\epsilon/4 & 0 \\ -\epsilon/4 & \epsilon/3 & 0 \\ 0 & 0 & 2\epsilon/3 \end{bmatrix}$$

b) Calculate the angular momentum vector $\vec{L}$.

$$\vec{\omega} = \omega(t) \hat{e}_3 \Leftrightarrow \vec{\omega} \leftrightarrow \begin{bmatrix} 0 \\ 0 \\ \omega(t) \end{bmatrix}$$

$$\vec{L} = [\vec{I}] \cdot \vec{\omega} = \begin{bmatrix} \frac{\epsilon}{3} & -\frac{\epsilon}{4} & 0 \\ -\frac{\epsilon}{4} & \frac{\epsilon}{3} & 0 \\ 0 & 0 & \frac{2\epsilon}{3} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \omega(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{2\epsilon \cdot \omega(t)}{3} \end{bmatrix}$$

$$L_3 = \frac{2\epsilon \theta(t)}{3}$$

c)